Flexible Cable Simulation in NVIDIA Isaac Sim

Comparing three modelling approaches for RL-based cable manipulation

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Outline

Motivation

- **Why cables?** wire-harness assembly, cable routing, and deformable-object manipulation are common, unsolved robotics tasks.
- **The challenge:** Isaac Sim has *no native 1-D deformable (rod/cable) body* – only rigid bodies and volumetric FEM soft bodies.
- **This study:** implement three candidate cable models and lay out *what each can and cannot do* – their capabilities and limitations – as a basis for choosing one later for cable manipulation.

Three approaches, each in two scenarios (free cable, and cable held by a Franka arm). This talk presents trade-offs and limitations – it does not pick a winner.

Three approaches

1. Capsule chain

rigid capsules + D6 joints; bending from Euler–Bernoulli beam theory (PhysX).

2. Cosserat rod (Warp)

1-D position-based elastic rod in NVIDIA Warp, after the *OGC* SIGGRAPH-2025 paper.

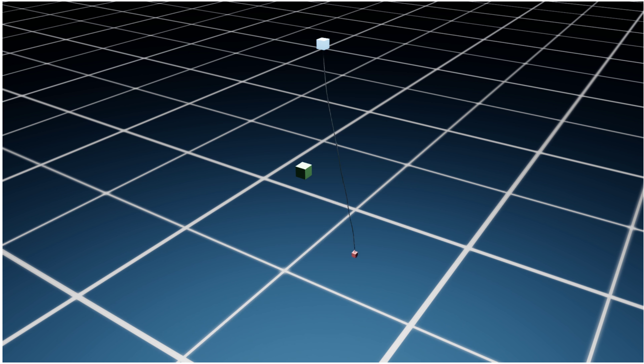
3. FEM deformable

Isaac Sim's built-in volumetric soft body (tetrahedral FEM).

Two scenarios each ⇒ **6 simulations**: (A) free hanging/swinging cable (B) cable connected to a robot manipulator.

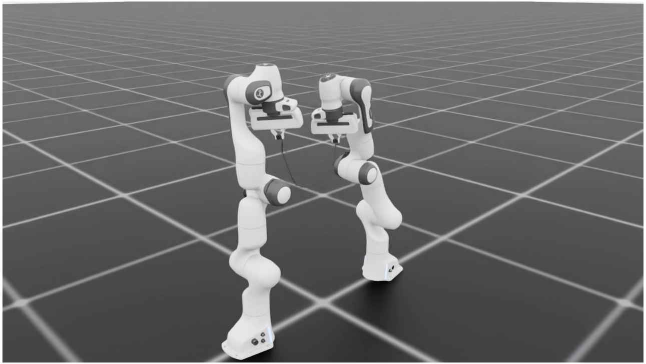
Approach 1 — Capsule chain (D6 joints)

- Rigid capsules linked by **D6 joints**; translations locked (inextensible), bending springs $k = EI/L$ from beam theory.
- PhysX **TGS** solver, up to 200 links, 240 Hz.
- Reproduces the **Govoni et al. 2025** stability sweep [2].
- Modes: hanging-kick and both-ends-fixed.



Video: 1_capsule_cable.mp4 (shared Drive folder)

Approach 1 — With robot

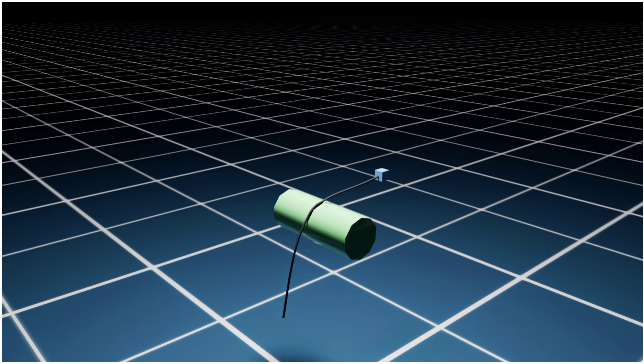


Video: 1_capsule_robot.mp4 (shared Drive folder)

Inextensible (rigid joints): a hard pull **drags the second arm ∼17 cm** while the cable barely stretches (∼5 cm) ⇒ **true force transmission**.

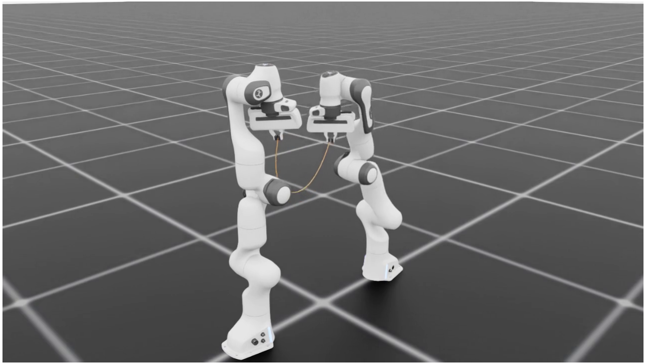
Approach 2 — Cosserat rod in NVIDIA Warp

- 1-D **Cosserat / XPBD** elastic rod: inextensible stretch + bending stiffness + gravity, solved on the GPU in **Warp**.
- Idea from the **OGC** paper (Chen et al., SIGGRAPH 2025) [1]: penetration-free contact for cloth & rods.
- Truly **thin** (no FEM fattening); **exact** node-vs-primitive collision.



Video: 2_cosserat_cable.mp4 (shared Drive folder)

Approach 2 — With robot

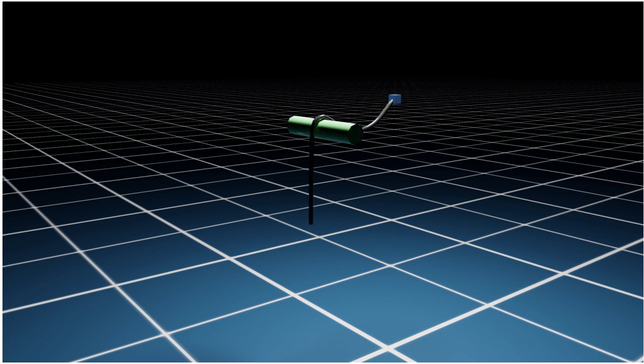


Video: 2_cosserat_robot.mp4 (shared Drive folder)

End nodes **pinned** to the grippers: the thin rod follows the arms and sags cleanly between them (best-looking cable).
One-way – it follows but exerts no force back, so it doesn't drag the second arm.

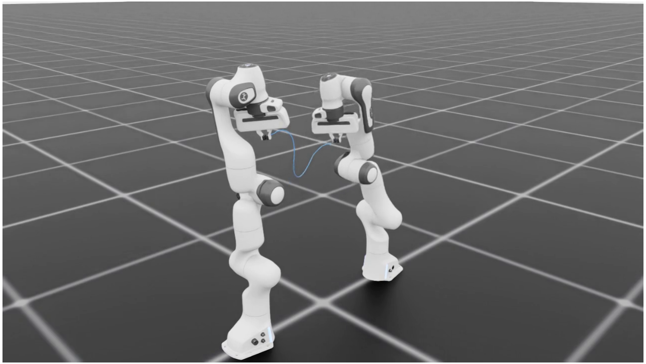
Approach 3 — FEM deformable (Isaac Sim)

- Isaac Sim’s built-in **volumetric** (tetrahedral) FEM soft body (NVIDIA Omniverse volume-deformable workflow, [3]).
- TPU material: $E = 40\,\mathrm{MPa}$, $\nu = 0.48$, $\rho = 1150\,\mathrm{kg/m^3}$.
- Rod **fattened** to 12 mm and E rescaled so it bends/weighs like the real 1.5 mm cable (voxeliser needs ≥ 3 hexes across).
- Rendered thin via a swept tube on its centreline.



Video: 3_fem_cable.mp4 (shared Drive folder)

Approach 3 — With robot



Video: 3_fem_robot.mp4 (shared Drive folder)

Region-weld attachment. Being a soft body it **stretches a lot** under a hard pull (~20–35%) and drags the follower little ⇒ poor force transmission.

Trade-offs at a glance

Criteria	**D6 capsule**	**Cosserat (Warp)**	**FEM**
Cable realism	beam-bending	1-D rod (thinnest)	volumetric (jelly/stretch)
Stretches?	no (rigid links)	no (inextensible)	yes (∼10–50%)
Thinness	true 1.5 mm	true (any radius)	≥12 mm (faked thin)
Compute cost	low	low (∼1.4× RT)	high (<0.2× RT)
Robot force coupling	two-way	one-way (or spring)	two-way (rough)
Collision	exact capsules	exact (coded prims)	approximate tets
Setup / code	many joints	custom Warp solver	built-in

RT = real time. These are *trade-offs* – no single method wins on every row; the right one depends on the priority.

Pros and cons (each is a trade-off — no single "best")

1. **Capsule chain (D6 joints)**
   - \+ Stable; physically grounded; real-time; two-way force coupling; easy to randomize.
   - – Piecewise-rigid (no axial give); many bodies; needs a heavy solver to stay inextensible.
2. **Cosserat rod in Warp (OGC)**
   - \+ Thinnest, cleanest cable; inextensible; no jelly; exact node-vs-primitive collision; GPU-fast.
   - – Separate (non-PhysX) sim; only artificial robot coupling; hand-coded collision; custom solver.
3. **FEM deformable**
   - \+ True volumetric soft body; native to Isaac Sim; two-way; handles squashing/draping.
   - – Fat, stretches, jelly, rigid at the joint, rough collision, slow.

Limitations — Capsule chain (D6 joints)

- **Piecewise-rigid:** links cannot stretch *at all* – it is a chain of rigid bodies, not a continuum; bending is a discrete per-joint spring.
- **Many bodies:** needs 60–200 links + joints for a smooth look $\Rightarrow$ many rigid bodies and constraints to simulate.
- **Solver cost vs. rigidity:** staying inextensible under load needs high position-iteration counts ($\approx$180); too few $\Rightarrow$ it stretches, too many $\Rightarrow$ it goes unstable.
- **Numerical twist:** the free twist DOF spins up unless explicitly damped.
- **Tuning-sensitive:** link mass / joint stiffness / damping / timestep must be balanced (mass-ratio & dt limits) or the chain diverges.
- **Approximate bending:** torsional springs from beam theory, not a true rod.

Limitations — Cosserat rod (Warp / OGC)

- **Not a native physics body:** it is a *separate* Warp simulation $\Rightarrow$ it does not interact with PhysX bodies through the engine (no automatic contact with robots or objects).
- **Artificial robot coupling:** either kinematic (**one-way** – follows the gripper, no force back) or a hand-tuned spring (two-way, but then it visibly *stretches*); no clean physical two-way coupling.
- **Hand-coded collision only:** node-vs-primitive (ground, sphere) that you write; no general mesh contact, no self-collision unless coded.
- **Implementation burden:** you must write & maintain the XPBD solver kernels – not off-the-shelf; every new obstacle needs custom contact code.

Limitations — FEM deformable

- **Cannot be thin:** voxeliser needs ≥3 hexes across the diameter and cooking caps resolution ⇒ thinnest ≈12 mm (real cable 1.5 mm, 8× too thick); only *rendered* thin.
- **Stretches:** one isotropic E ties bending ($\propto Er^4$) to axial ($\propto Er^2$) – exact bending ⇒ axially soft (~50% stretch); firm enough not to stretch ⇒ bending 20–30× too stiff.
- **"Jelly":** low-resolution soft body + near-incompressible ν ⇒ volumetric locking / wobble.
- **Rigid at the connection:** attachment *welds a region* of vertices (clamped-beam boundary) ⇒ straight/stiff near the joint, not a pin.
- **Rough collision:** simplified, re-meshed coarse collision tets ⇒ imprecise, can clip thin obstacles; self-collision off by default.
- **Expensive:** GPU FEM runs <0.2× real-time.

Full derivations: `why_fem_cant_be_thin.py`, `README_fem_attachment_collision.md`.

References

[1] A. H. Chen, J. Hsu, Z. Liu, M. Macklin, Y. Yang, and C. Yuksel. **Offset Geometric Contact.** *ACM Transactions on Graphics* (SIGGRAPH 2025), 44(4), 2025. DOI: 10.1145/3731205.
Source for Approach 2 (Cosserat / position-based rod in NVIDIA Warp). `graphics.cs.utah.edu/research/projects/ogc`

[2] A. Govoni, N. Zubair, S. Soprani, and G. Palli. **Performance Analysis of a Mass–Spring–Damper Deformable Linear Object Model in Robotic Simulation Frameworks.** European Robotics Forum (ERF), 2025. arXiv:2504.13659.
Reference for the Approach 1 DLO / stability behaviour.

[3] NVIDIA. **Omniverse Physics Extension — Volume Deformables (Overview).** Omniverse tutorial video. `youtube.com/watch?v=focL232tb8U`
Source for Approach 3 (Isaac Sim volumetric FEM soft body).

Thank you

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